

Placement in OpenROAD

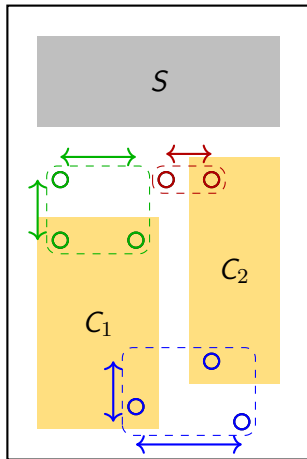
Placement

Input:

- Chip area $[x_{\min}, x_{\max}] \times [y_{\min}, y_{\max}]$
- Set of rectangular cells $\mathcal{C}$
- Set of rectangular blockages $\mathcal{S}$
- Set of pins P placed on the chip area or on cells
- Partition $\mathcal{N}$ of P into nets, net weights $w : \mathcal{N} \rightarrow \mathbb{R}^+$

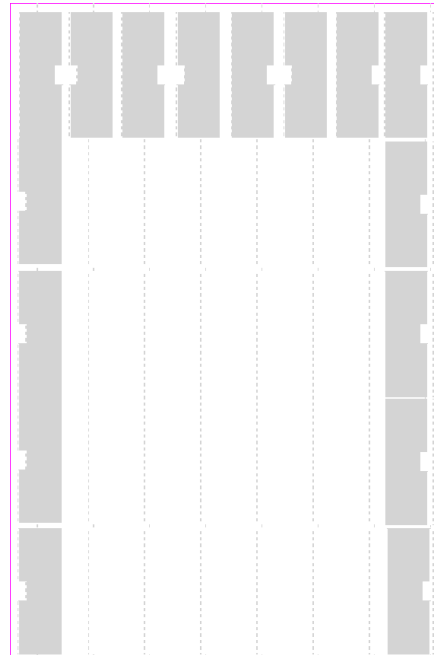
Task:

- Find a packing of $\mathcal{C}$ into chip area
- with $\mathcal{C} \cup \mathcal{S}$ overlap-free
- minimizing weighted bounding box netlength



Placement

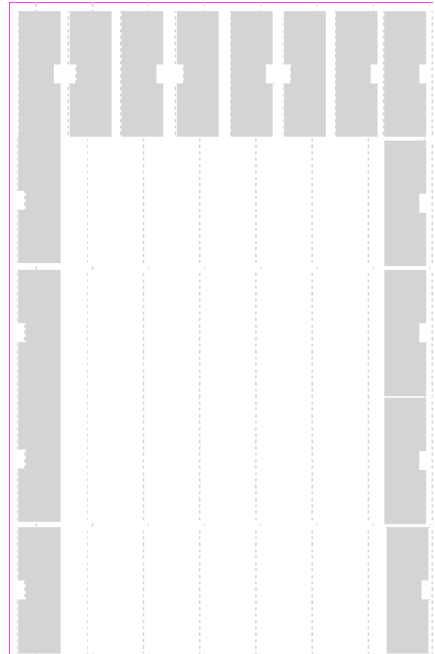
Input:



Placement

Input:

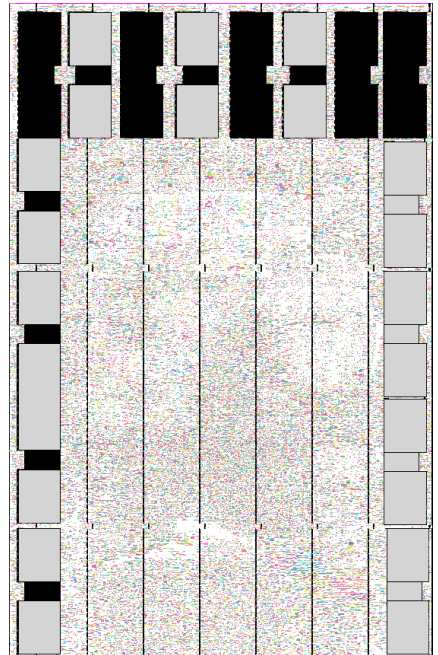
- chip area with blockages



Placement

Input:

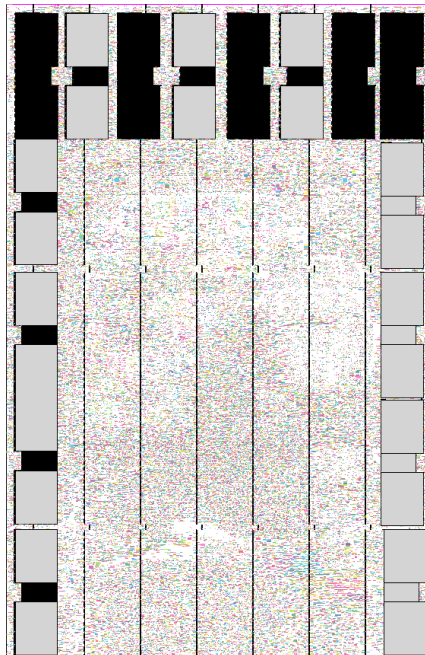
- chip area with blockages
- cells



Placement

Input:

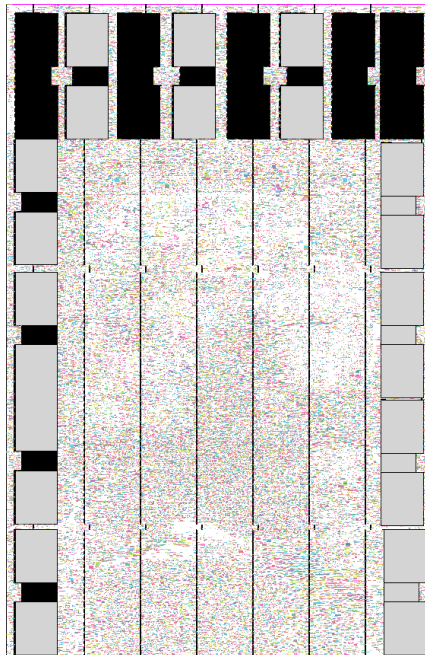
- chip area with blockages
- cells
- nets



Placement

Input:

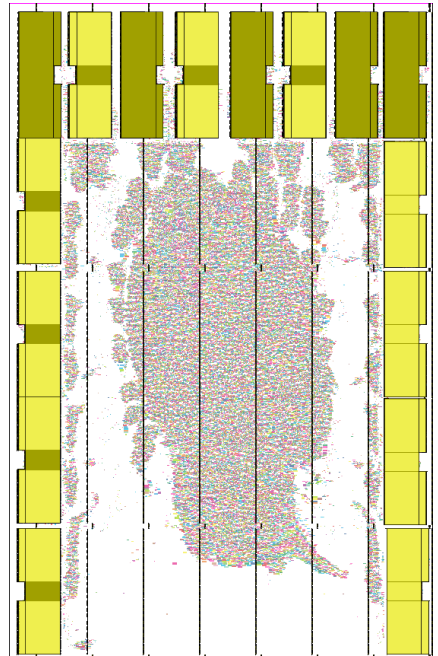
- chip area with blockages
- cells
- nets



Placement

Input:

- chip area with blockages
- cells
- nets



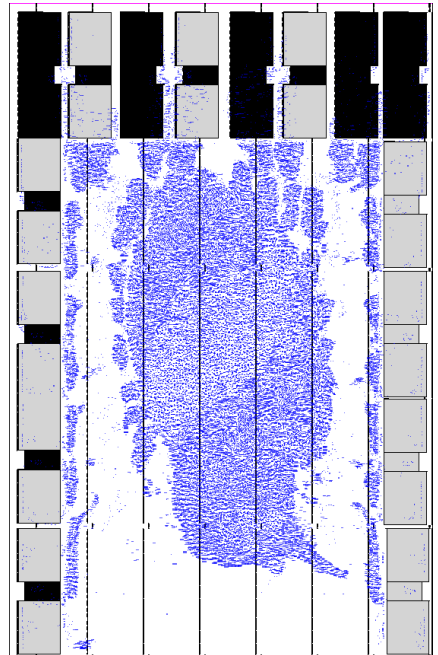
Placement

Input:

- chip area with blockages
- cells
- nets

Global Placement

- allow local overlaps, no large overfull regions



Placement

Input:

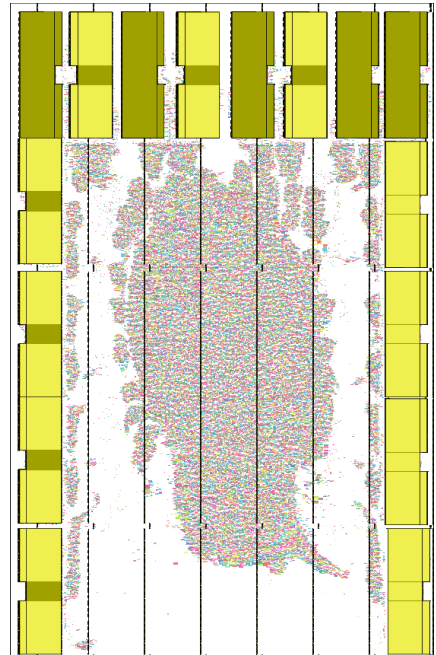
- chip area with blockages
- cells
- nets

Global Placement

- allow local overlaps, no large overfull regions

Legalization (Detailed Placement)

- remove overlaps, do not change too much



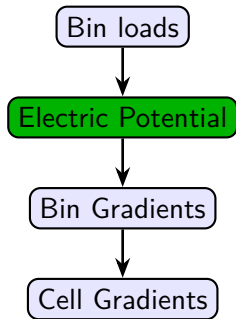
Electrostatic Global PLacement

- Minimizing $\text{wavelength} + \lambda \cdot \text{density_penalty}$
 - ▶ **Wavelength**: Differentiable approximation of HPWL
 - ▶ **Density penalty**: Model cells as electric charges, minimize potential energy
- Efficient density computation (FFT) introduced by ePlace [Lu et al. 2015]
- Nesterov accelerated gradient descent
- State-of-the-art approach for global placement

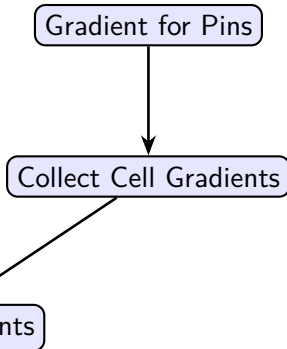


Basic gradient computation flow

Density



Wirelength

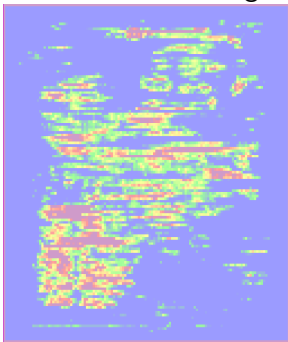


Electric potentials

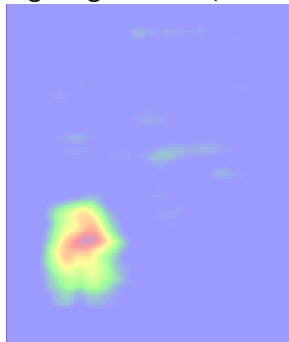
Input:

- Bin set I with graph structure (grid)
- Bin deltas $\rho_i = \text{load}_i - \text{target}_i$ with $\sum_{i \in I} \rho_i \leq 0$

Output: Potential measuring overcrowdedness, considering neighbours N_i

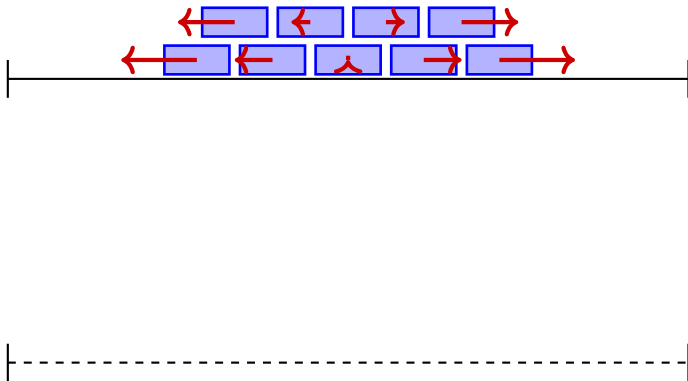


Bin deltas

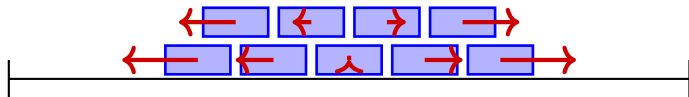


Potential

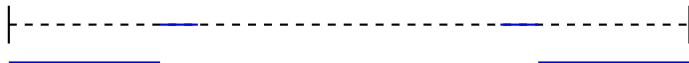
Electric potential and field



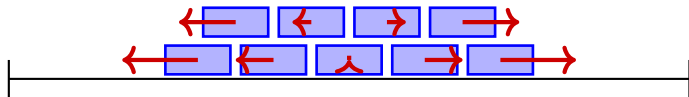
Electric potential and field



charge density

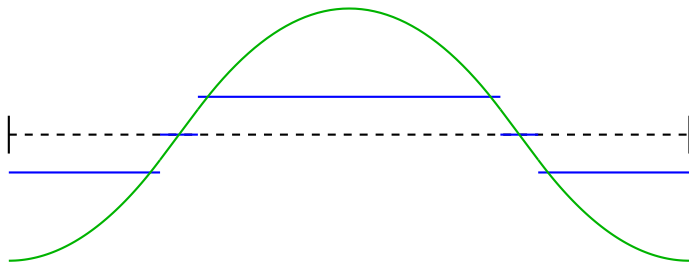


Electric potential and field

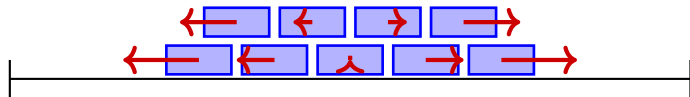


potential

charge density



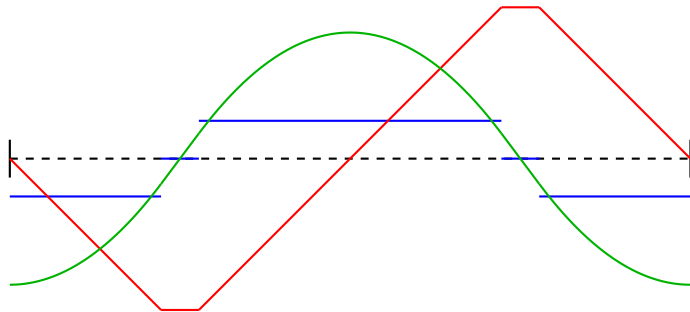
Electric potential and field



potential

electric field

charge density



Filler cells

- Electric fields always aim for equal distribution
- in practice:
 - ▶ target density
 - ▶ more space than needed

Filler cells

- Electric fields always aim for equal distribution
- in practice:
 - ▶ target density
 - ▶ more space than needed

⇒ add filler cells to consume unused space

Legalization in OpenROAD

- Remove overlaps minimizing average and maximum displacement
- Allow mixed-sized cells
- Consider movebounds (fence regions)
- Power alignment

Negotiation-based legalization

- **Idea:** Rip-up and re-route for placement
- Cells can be placed at grid locations

Negotiation-based legalization

- Idea: Rip-up and re-route for placement
- Cells can be placed at grid locations
- Iterate:
 - ▶ Increase cost (based on history and current usage) on overful grid locations
 - ▶ Move cells to cheaper location

Negotiation-based legalization

- **Idea:** Rip-up and re-route for placement
- Cells can be placed at grid locations
- Iterate:
 - ▶ Increase cost (based on history and current usage) on overful grid locations
 - ▶ Move cells to cheaper location
- **Postoptimization:**
 - ▶ Iteratively move to cheaper free locations
 - ▶ Matching-based cell-swapping

OpenROAD Commands

- `global_placement [options]`
- `detailed_placement [options]`

Videos



Clickable link